

Topic 1

Number

**Concept guide · Worked examples · Practice · Investigation and modelling
· Solutions****Learning focus**

Content explanations, original Cambridge-style practice and an investigation task tailored to International Mathematics 0607.

Topic map

1.1	Types of number, factors and multiples
1.2	Sets and Venn diagrams
1.3	Powers, roots and standard form
1.4	Fractions, decimals and percentages
1.5	Ratio, rates, proportion and money
1.6	Accuracy, bounds, indices and surds
1.7	Multiplicative models

Core foundations and Extended applications are integrated. Practise exact methods, calculator work and the written reasoning required in investigations.

1. Number structure

A prime number has exactly two positive factors. Prime factorisation writes an integer as a product of primes; the highest common factor uses the lowest common powers and the lowest common multiple uses the highest. A rational number can be written as a fraction of integers; irrational numbers cannot.

Worked example

For $24 = 2^3 \times 3$ and $36 = 2^2 \times 3^2$, $\text{HCF} = 2^2 \times 3 = 12$ and $\text{LCM} = 2^3 \times 3^2 = 72$.

Exam check: One is neither prime nor composite.

2. Sets and regions

A universal set contains all objects under discussion. $A \cap B$ is the overlap, $A \cup B$ includes either set, and A' is the complement relative to the universal set. Fill the overlap first, then the exclusive parts, then the outside region.

Worked example

Of 30 pupils, 18 take French, 14 take Spanish and 6 take both. French only = 12; Spanish only = 8; neither = $30 - (12 + 6 + 8) = 4$.

Exam check: Add the four separate regions once each; do not add both set totals without removing the overlap.

3. Fractions and percentage change

To add fractions, use a common denominator. A percentage change is change divided by the original amount. Successive changes multiply factors, so an increase of 20% followed by a decrease of 20% gives $1.2 \times 0.8 = 0.96$, a 4% overall decrease.

Worked example

A coat costs \$68 after a 15% discount. The final price is 85% of the original, so original = $68/0.85 = \$80$.

Exam check: In reverse percentages divide by the multiplier; $\$68 \div 0.15$ is not the original price.

4. Ratio, rates and proportionality

A ratio compares parts in matching units. For 2:3, the total is five equal parts. Convert compound units before using a speed, density or currency rate. Direct proportion has $y = kx$; inverse proportion has $y = k/x$.

Worked example

A map scale of 1:25000 makes 4.8 cm on the map equal to 120000 cm = 1.2 km on the ground.

Exam check: Always state the unit after converting a scale or a compound rate.

5. Standard form, indices and surds

Standard form is $a \times 10^n$ with $1 \leq |a| < 10$. Use $a^m \times a^n = a^{m+n}$ and $(a^m)^n = a^{mn}$. Simplify a surd by extracting square factors; rationalise a denominator when asked.

Worked example

$$(3 \times 10^4)(2 \times 10^{-3}) = 6 \times 10^1 = 60; \sqrt{72} = \sqrt{(36 \times 2)} = 6\sqrt{2}.$$

Exam check: Negative powers mean reciprocals, not negative values.

6. Rounding and bounds

If a length is rounded to the nearest 0.1 cm, its exact value is at least measured value - 0.05 and less than measured value + 0.05. For positive factors, use both lower bounds for the least product and upper bounds for the greatest. Round only at the end.

Worked example

A length recorded as 8.4 cm to the nearest 0.1 cm has $8.35 \leq L < 8.45$ cm.

Exam check: The upper boundary is excluded when values exactly at it round up.

7. Multiplicative models

Percentage growth is modelled by initial value times $(1+r)^n$ after n equal time steps; decay uses $1-r$. Compare successive ratios in a table before choosing an exponential model. State whether whole-number rounding is sensible in context.

Worked example

An account with \$400 earning 5% annually has model $A(n)=400(1.05)^n$. After 2 years $A(2)=\$441$.

Exam check: An interest rate or population growth rate may not remain constant forever.

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Graphical calculator and Paper 6

Use exact fraction and standard-form entries; check order of magnitude before accepting a decimal. A multiplicative growth model can be explored in a table.

State the rule or model, display enough working to reproduce your result, test it against the given values and discuss at least one limitation.



Practice paper

Topic 1 · Number

Instructions

Original practice, informed by June 2026 components. Section A: short multiple choice for revision without a calculator. Section B: show full written working. Section C: investigation and modelling with reasons and checks.

Candidate name:

Class / date:

Total: 40 marks

Section A · Multiple choice

Choose one answer for each question.

1. Which is prime? [1]

- A 1
- B 21
- C 29
- D 39

2. What is HCF(18,30)? [1]

- A 3
- B 6
- C 9
- D 90

3. What is 15% of 80? [1]

- A 8
- B 12
- C 15
- D 20

4. Write 0.00056 in standard form. [1]

- A 5.6×10^{-4}
- B 5.6×10^{-5}
- C 56×10^{-5}
- D 0.56×10^{-3}

5. What is $\frac{2}{3} + \frac{1}{4}$? [1]

- A $\frac{3}{7}$
- B $\frac{11}{12}$
- C $\frac{5}{12}$
- D $\frac{8}{12}$

6. What is the decimal multiplier for a 12% reduction? [1]

- A 0.12
- B 1.12
- C 0.88
- D 0.92

7. For a ratio 3:5, what fraction is the first part? [1]

- A $\frac{3}{5}$
- B $\frac{5}{8}$
- C $\frac{3}{8}$
- D $\frac{8}{3}$

8. Which is the lower bound of 6.2 to the nearest 0.1? [1]

- A 6.1
- B 6.15
- C 6.20
- D 6.25

Section B · Structured questions

9. A shop reduces a \$90 jacket by 20%, then increases the reduced price by 10%.

(a) Find the price after the reduction.

[2]

(b) Find the final price.

[2]

(c) Find the overall percentage change from \$90.

[3]

[Total: 7]

10. Two buses depart together at 09:00. One repeats every 18 minutes and the other every 30 minutes.

(a) Write 18 and 30 as products of prime factors.

[2]

(b) Find the least common multiple.

[2]

(c) State the next time both buses depart together. [2]

[Total: 6]

11. In a group of 40, 23 study Art, 18 study Music and 8 study both.

(a) Find the number studying only Art. [1]

(b) Find the number studying neither subject. [2]

(c) Find the probability of selecting a pupil studying exactly one subject. [2]

[Total: 5]

12. A rectangular garden has sides 7.4 m and 5.6 m, each measured to the nearest 0.1 m.

(a) Write the lower bound for the longer side. [1]

(b) Calculate the upper bound for the perimeter.

[2]

(c) Calculate the lower bound for the area.

[3]

[Total: 6]

Section C · Investigation and modelling

Show the rule, test it with values and explain the result in context.

13. A club has 200 members and its membership grows by 10% each year.

(a) Calculate the number predicted after one and two years. [2]

(b) Write an expression $M(n)$ for n years. [2]

(c) Use the model to predict after five years, to the nearest whole member. [2]

(d) State one reason the prediction could fail over many years. [2]

[Total: 8]



Worked solutions

Fold back this section while completing the practice paper.

Marking routine

Method, model, communication and verification matter. A numerical answer should follow from clear working.

Section A · Answers and reasons

1. C — 29. 29 has no positive divisors other than 1 and 29.
2. B — 6. $18 = 2 \times 3^2$ and $30 = 2 \times 3 \times 5$; their common factor is 2×3 .
3. B — 12. $0.15 \times 80 = 12$.
4. A — 5.6×10^{-4} . Move the decimal four places right to obtain 5.6.
5. B — $11/12$. Use denominator 12: $8/12 + 3/12 = 11/12$.
6. C — 0.88. Retain $100\% - 12\% = 88\%$, giving multiplier 0.88.
7. C — $3/8$. The first part is 3 of $3 + 5 = 8$ equal parts.
8. B — 6.15. The half-unit is 0.05; $6.2 - 0.05 = 6.15$.

Section B • Worked solutions

9. a) $\$90 \times 0.8 = \72 [2]. b) $\$72 \times 1.1 = \79.20 [2]. c) $(79.20 - 90)/90 \times 100 = 12\%$ decrease [3].

10. a) $18 = 2 \times 3^2$; $30 = 2 \times 3 \times 5$ [2]. b) $\text{LCM} = 2 \times 3^2 \times 5 = 90$ min [2]. c) $09:00 + 90 \text{ min} = 10:30$ [2].

11. a) $23 - 8 = 15$ [1]. b) Music only = 10, union = $15 + 8 + 10 = 33$, neither = 7 [2]. c) Exactly one = $15 + 10 = 25$, so $25/40 = 5/8$ [2].

12. a) 7.35 m [1]. b) $2(7.45 + 5.65) = 26.2$ m (strictly less than this bound) [2]. c) $7.35 \times 5.55 = 40.7925 \text{ m}^2$ [3].

Section C • Investigation and modelling solutions

13(a). 220 then 242 [2].

13(b). $M(n) = 200(1.1)^n$ [2].

13(c). $M(5) = 322.102$, so about 322 members [2].

13(d). Recruitment conditions or capacity can change; a fixed percentage each year is an assumption [2].

Self-check

Explain why your model fits the starting data, show an independent check and name a condition under which your prediction would be unreliable.

Reference: Cambridge IGCSE International Mathematics 0607 syllabus 2025-2027, <https://www.cambridgeinternational.org/Images/662472-2025-2027-syllabus.pdf>. Assessment conventions informed by the June 2026 papers and mark schemes supplied by the user. All practice questions are newly written.